Evaluating Cost-Sensitive Machine Learning Models for Credit Default Prediction

Reducing Financial Risk using Machine Learning

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Course : Msc Data Science Project

Github Link : [Credit-default-cost-sensitive ml]

Background

The Credit Decision Challenge

Banks issue credit cards to customers and must predict whether each customer will repay their debt or default on payments. This decision directly impacts the bank's financial health and profitability.

The Risk of Misclassification

When a bank wrongly approves a high-risk customer, it results in substantial financial loss through unpaid debts and collection costs. Machine learning has become essential for automating and improving these critical credit decisions.

Problem Definition

False Positive

Rejecting a good customer who would have repaid

Opportunity cost

False Negative

Approving a risky customer who defaults

Direct financial loss

In credit prediction, false negatives are significantly more dangerous because they cause immediate financial loss through unpaid debt, whereas false positives represent lost opportunity.

Limitation of Traditional Models

Standard Approach

Most machine learning models optimise for **accuracy** as the primary metric

Hidden Assumption

Accuracy treats all classification errors equally, assuming false positives and false negatives have identical costs

Real Banking Reality

False negative cost is substantially higher than false positive cost in financial institutions

Research Question

"How effective are cost-sensitive machine learning models in reducing financial risk compared to standard models in credit default prediction?"

Dataset

UCI Credit Card Default Dataset

- 30,000 credit card customers
- Country: Taiwan
- Year: 2005
- 24 features including demographics, credit limit, payment history, and bill amounts

Target Variable

`'default payment next month'`

- **1** = Default
- **0** = Non-default

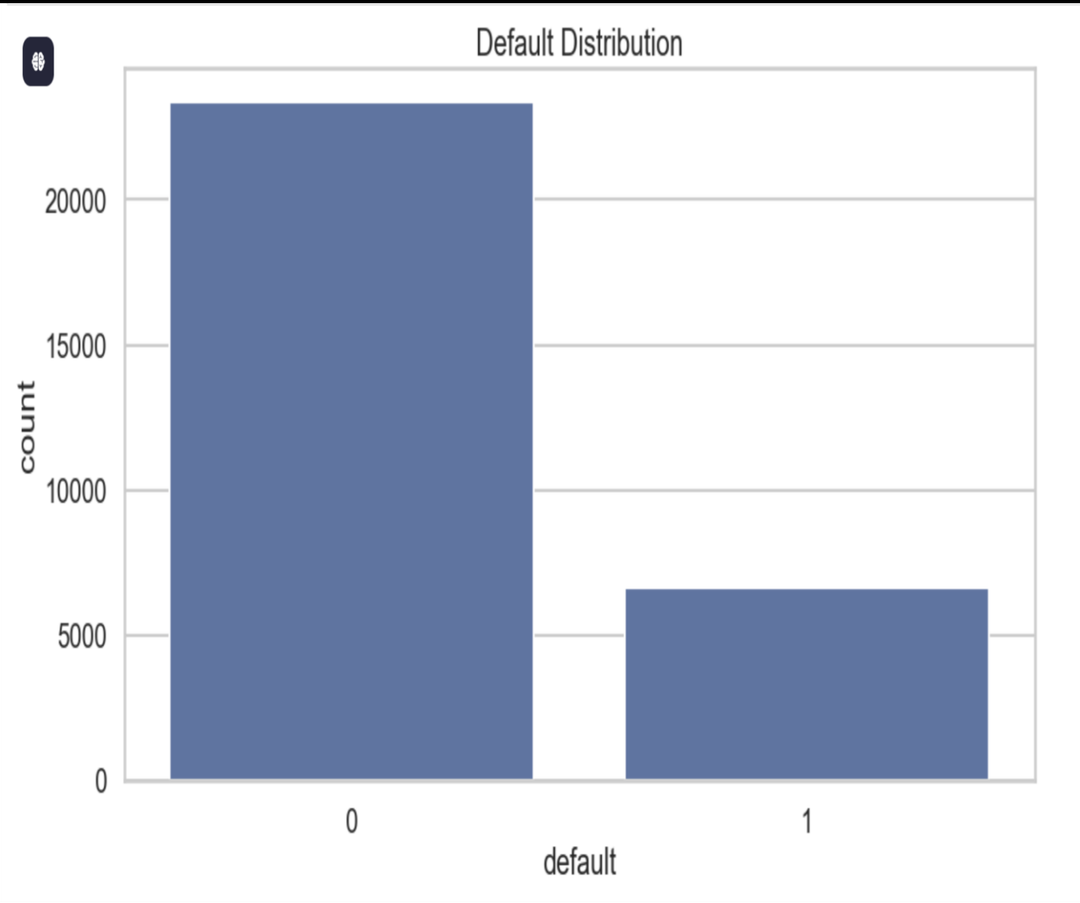
Key Characteristics

The dataset represents a real-world credit scoring scenario with historical payment behaviour and demographic information.

Class Imbalance

23,364	6,636	22.1%
Non-Default	Default	Default Rate
Majority class (0)	Minority class (1)	Imbalance ratio

The significant class imbalance means that models tend to predict the majority class and systematically ignore the minority defaulters, leading to poor detection of high-risk customers.



Feature Overview

LIMIT_BAL

Credit limit assigned to the customer

BILL_AMT1–6

Monthly bill amounts over 6 months

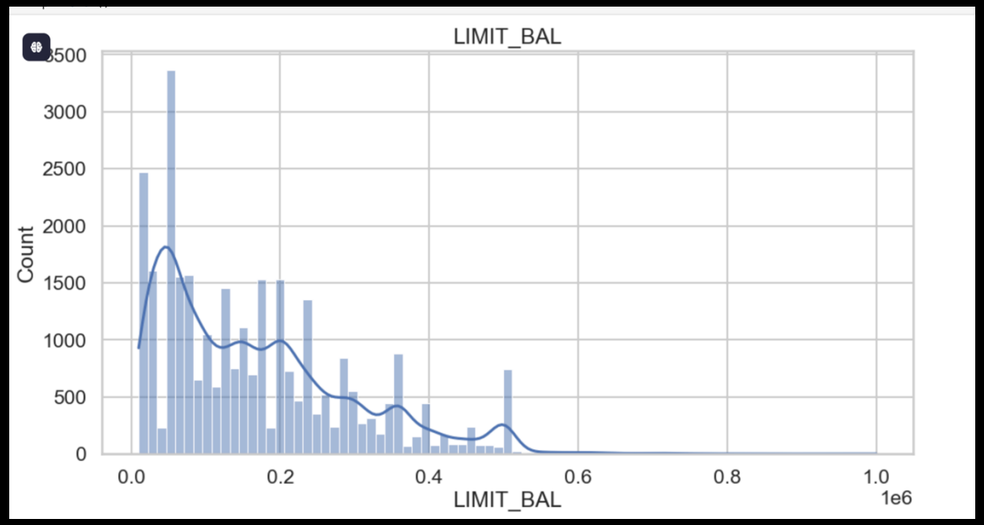
PAY_AMT1–6

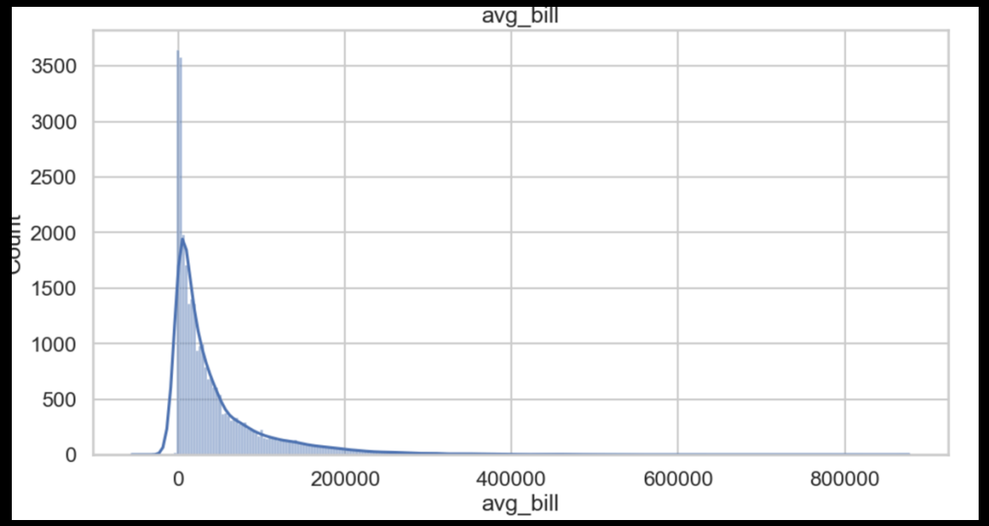
Payment amounts made over 6 months

PAY_0

Repayment status (most recent month)

Payment behaviour and bill payment patterns are critical predictors of default risk, as they directly reflect the customer's financial discipline and ability to meet obligations.





Data Exploration

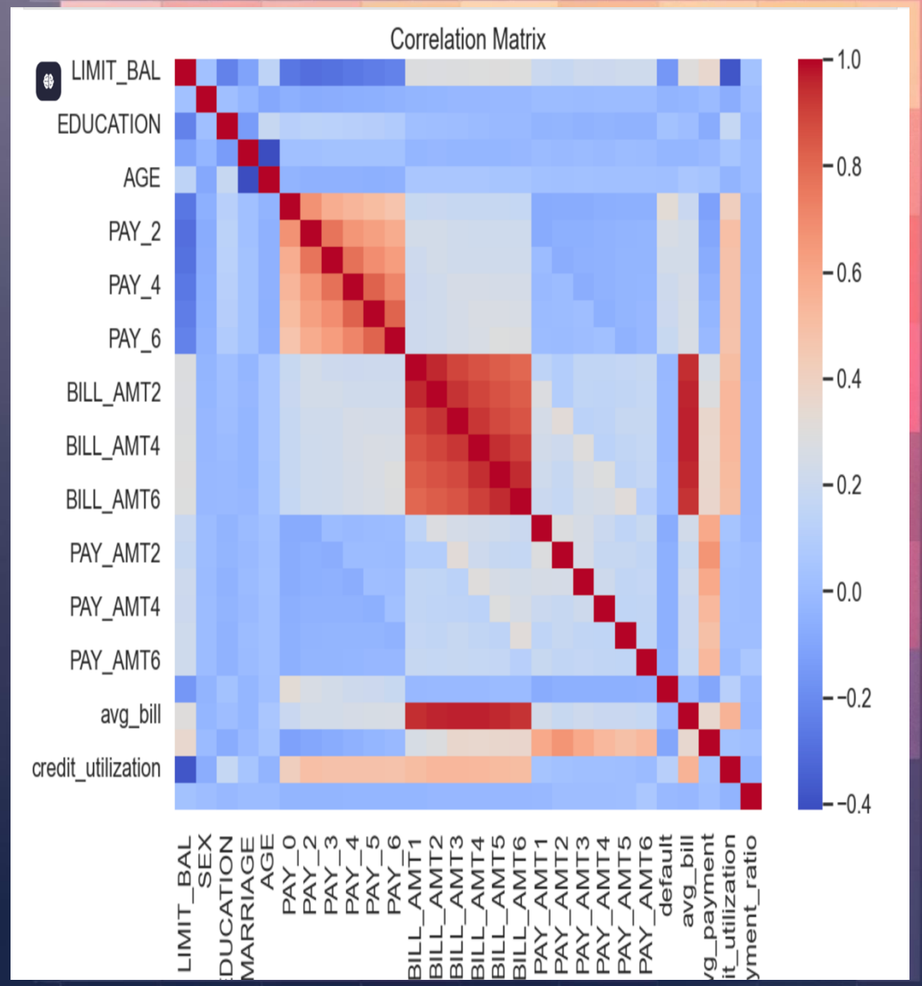
LIMIT_BAL Distribution

Credit limits show a right-skewed distribution with most customers having moderate limits, while a small number have very high credit limits.

Average Bill Distribution

Monthly bill amounts are heavily skewed, with some customers having exceptionally high usage patterns that may indicate higher risk.

Financial features exhibit substantial skewness, requiring appropriate transformation or robust modelling techniques to handle extreme values effectively.



Correlation Analysis

BILL_AMT Variables

Monthly bill amounts show high positive correlation across consecutive months, indicating consistent spending patterns

PAY_0 Correlation

Most recent repayment status demonstrates strong relationship with default outcome, making it a powerful predictor

Payment History

Past payment amounts show moderate correlation with default, providing valuable behavioural insights

Data Preprocessing Steps

01

Column Renaming

Target column renamed to 'default' for simplicity

02

EDUCATION Column

Invalid values (0, 5, 6) grouped into 'others' category

03

MARRIAGE Column

Missing value (0) treated as 'others'

04

Missing Values

Replaced using median values for avg_bill and credit_utilization

Feature Engineering

Creating meaningful features that capture customer behaviour

avg_bill

Average of BILL_AMT1–6

avg_payment

Average of PAY_AMT1–6

credit_utilization

avg_bill / LIMIT_BAL

payment_ratio

avg_payment / avg_bill

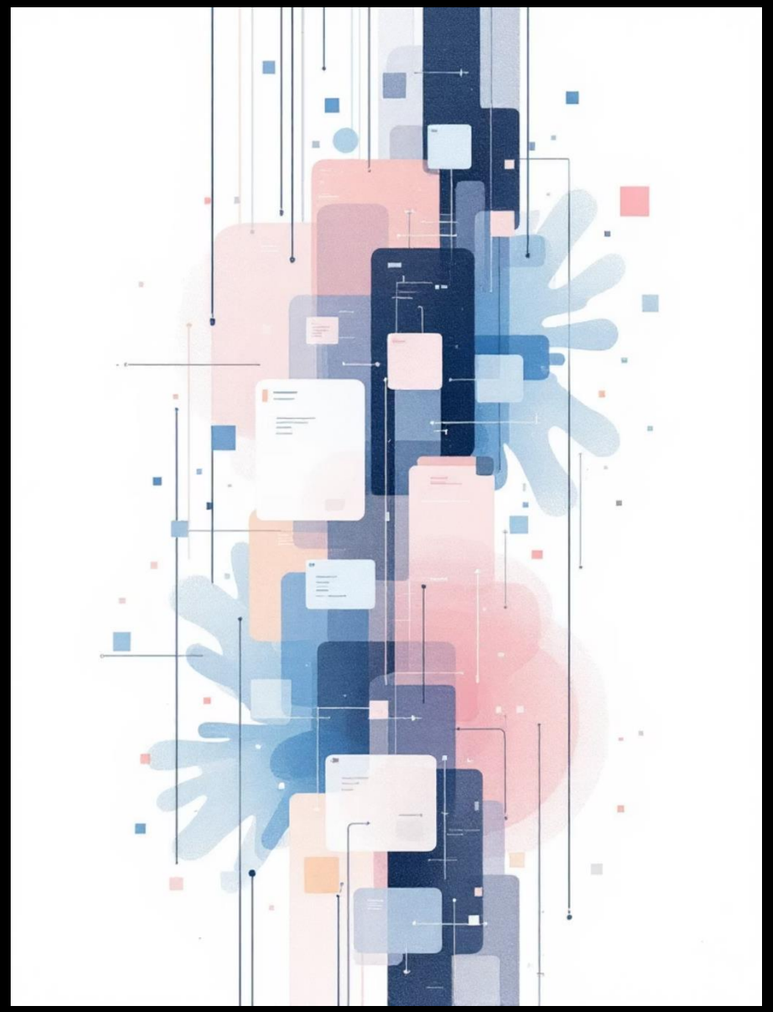
Data Splitting Strategy

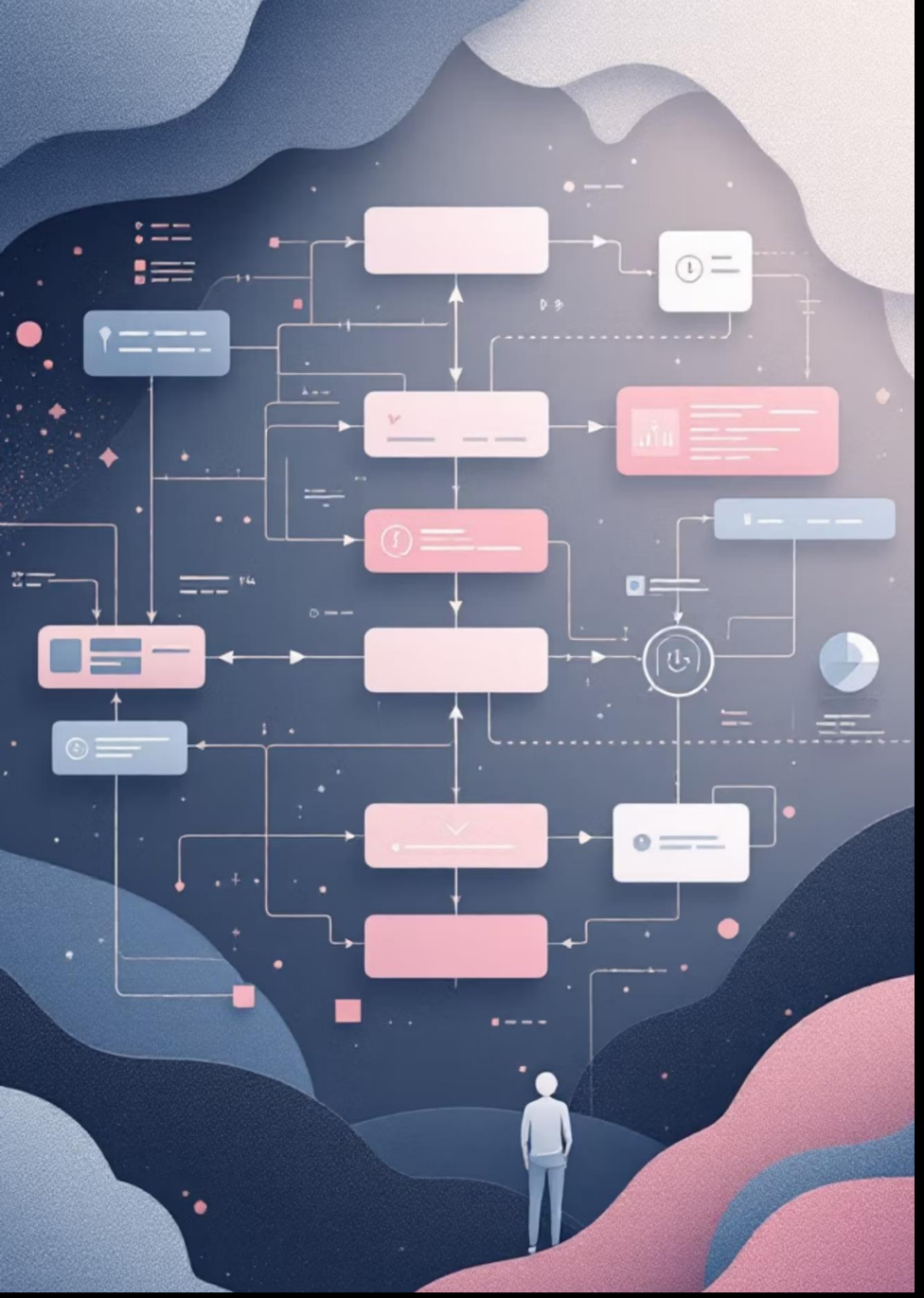
Method

- Train-test split: 80% / 20%
- Stratified sampling used

Why?

Maintains same ratio of defaulters in both training and test sets





Feature Scaling

StandardScaler Applied

Used with Logistic Regression for consistent feature scales

Not Applied to Tree Models

Random Forest and XGBoost do not require feature scaling

Models Employed

Logistic Regression

Baseline model for comparison

Random Forest

Handles non-linear relationships

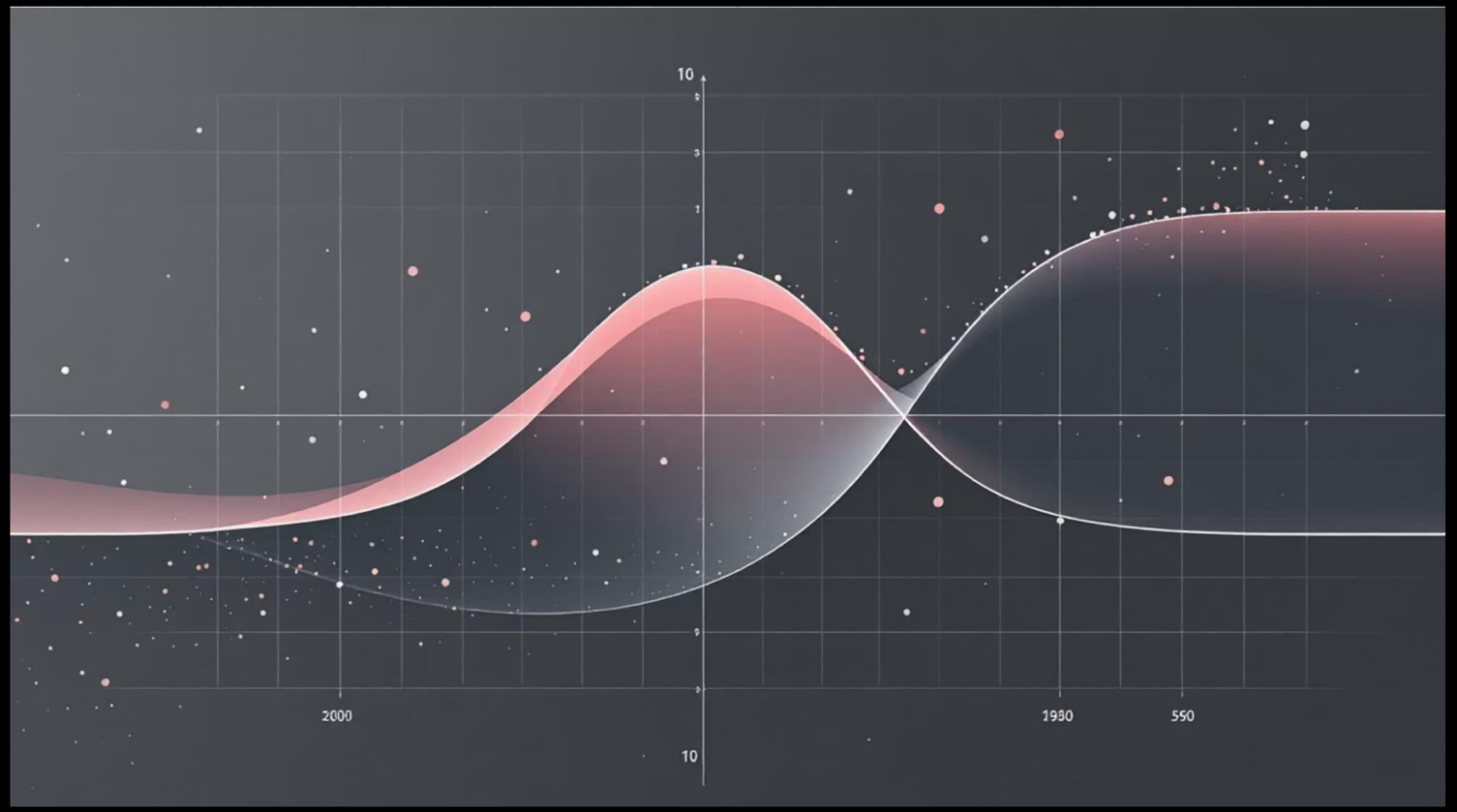
XGBoost

Powerful boosting algorithm

Initial Baseline Approach

Standard Logistic Regression

The initial model was trained using conventional logistic regression without any adjustments for class imbalance. This represents the baseline approach commonly used in practice.



Training Process

Model trained on scaled features with default parameters

Key Limitation

Equal treatment of both classes ignores real-world imbalance

Introducing Class Weights

Class 0 (Non-Default)

Weight: **0.64**

Lower penalty for misclassifying non-defaulters

Class 1 (Default)

Weight: **2.26**

Higher penalty for missing actual defaulters

This modification forces the model to penalise mistakes on defaulters more heavily, adjusting the loss function to reflect the business reality where missing a defaulter is significantly more costly than a false alarm.

Financial Cost Framework

Real-World Banking Risk

Rather than relying solely on accuracy, a custom financial cost function was developed to reflect actual banking losses.

₹10K

False Negative

Missed defaulter cost

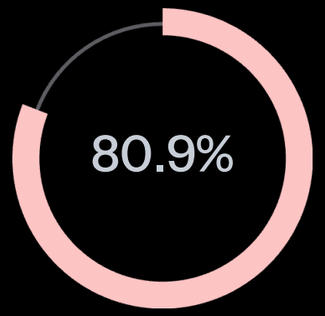
₹1K

False Positive

Incorrect flag cost

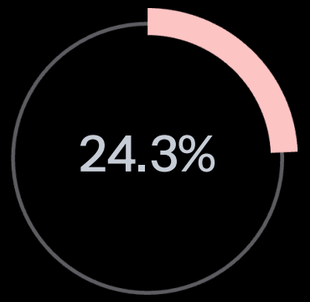
This approach shifts evaluation focus from statistical metrics to minimising actual financial loss, providing more meaningful business insights.

Baseline Model Performance



Accuracy

Correct predictions overall



Recall

Defaulters detected

What the Metrics Reveal

- Model correctly predicts many non-defaulters (high accuracy)
- Fails to identify most defaulters (low recall)
- Bias towards majority class remains

Interpretation

The model is over-predicting non-defaults due to class imbalance, missing critical risk cases that matter most to banks.

Cost-Sensitive Model Improvement

1

Retrained Model

Logistic regression with class weights applied

2

Modified Loss

Focus shifted to defaulter detection

3

Improved Recall

0.243 → 0.616

153%

Recall Increase

Compared to baseline

42%

False Negatives

Reduction achieved

The model now detects significantly more defaulters, substantially reducing false negatives and associated financial risk.

Comparative Model Analysis

36.5%

Random Forest

Trained on original data

56.6%

XGBoost

With scale_pos_weight

61.6%

Logistic Regression (CS)

With class weights

Key Findings

- Logistic regression responds best to class weighting
- XGBoost handles imbalance effectively
- Random Forest less sensitive to cost adjustments

Financial Cost Calculation

Confusion Matrix Extraction

False Negatives (FN) and False Positives (FP) extracted from predictions

Cost Formula Application

Cost = (FN × ₹10,000) + (FP × ₹1,000)

₹10.18M

Baseline Cost

Standard model

₹6.47M

Cost-Sensitive

Class-weighted

36.4%

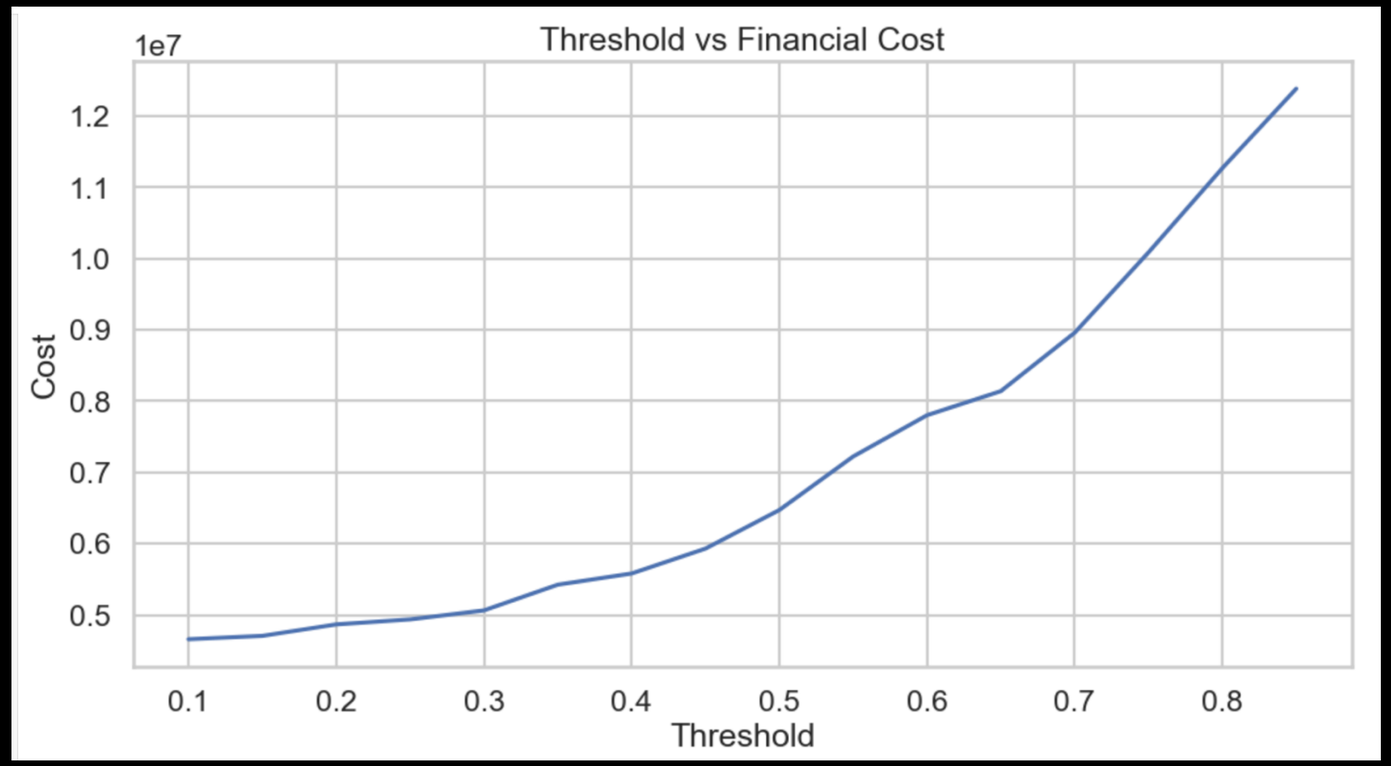
Cost Reduction

Financial savings

Threshold Optimisation Strategy

Dynamic Threshold Tuning

Instead of using the default threshold (0.5), multiple thresholds from 0.1 to 0.9 were tested. For each threshold, predictions were generated and financial cost calculated.



01

Generate Predictions

At varying thresholds

02

Calculate Costs

Using financial formula

03

Identify Minimum

Optimal threshold found

0.1

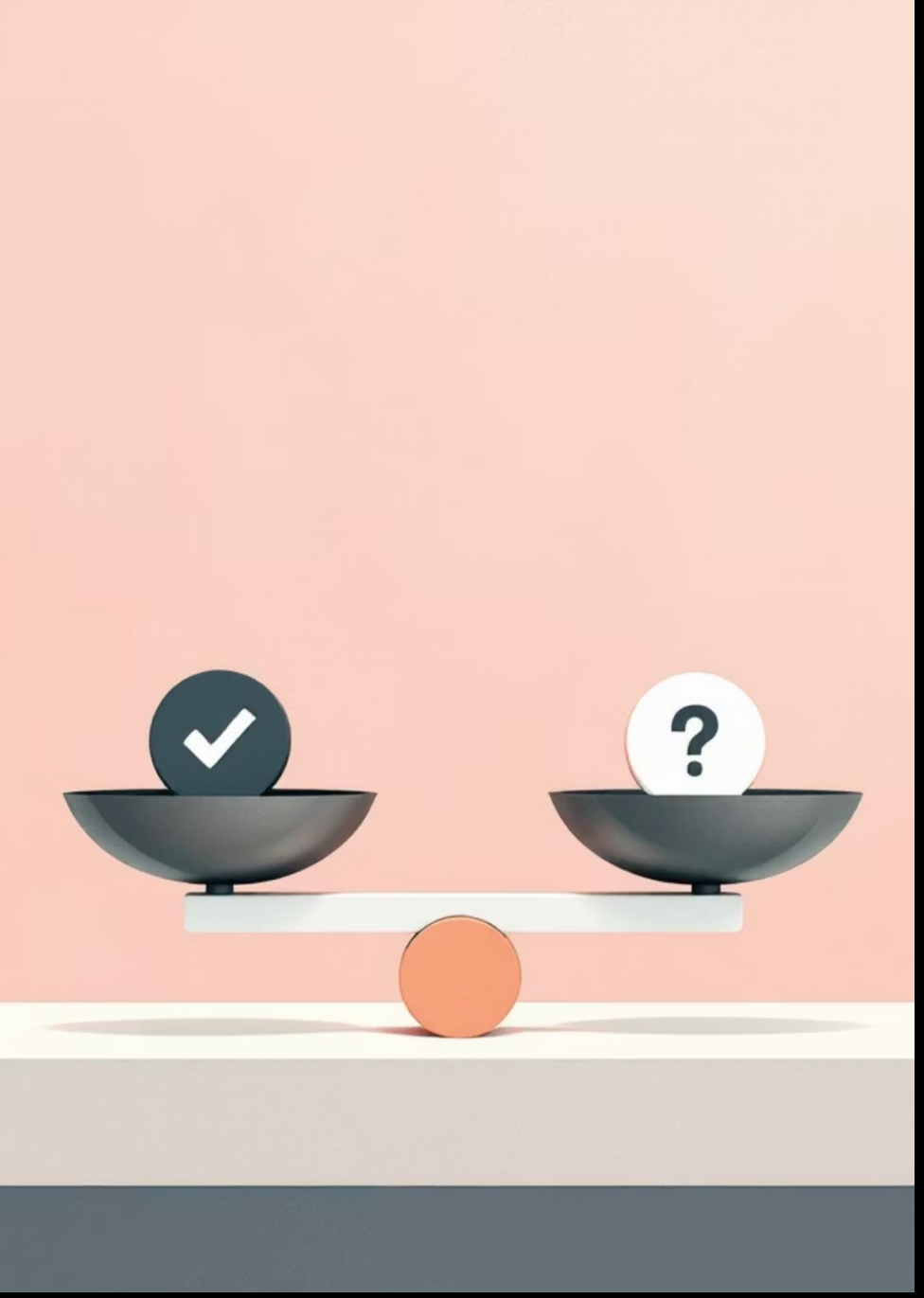
Optimal Threshold

Lower than default

₹4.66M

Minimum Cost

Further reduction



Why Threshold Matters

Default Threshold Limitation

Threshold of 0.5 designed for balanced problems, misses minority class in imbalanced scenarios

Lower Threshold Benefit

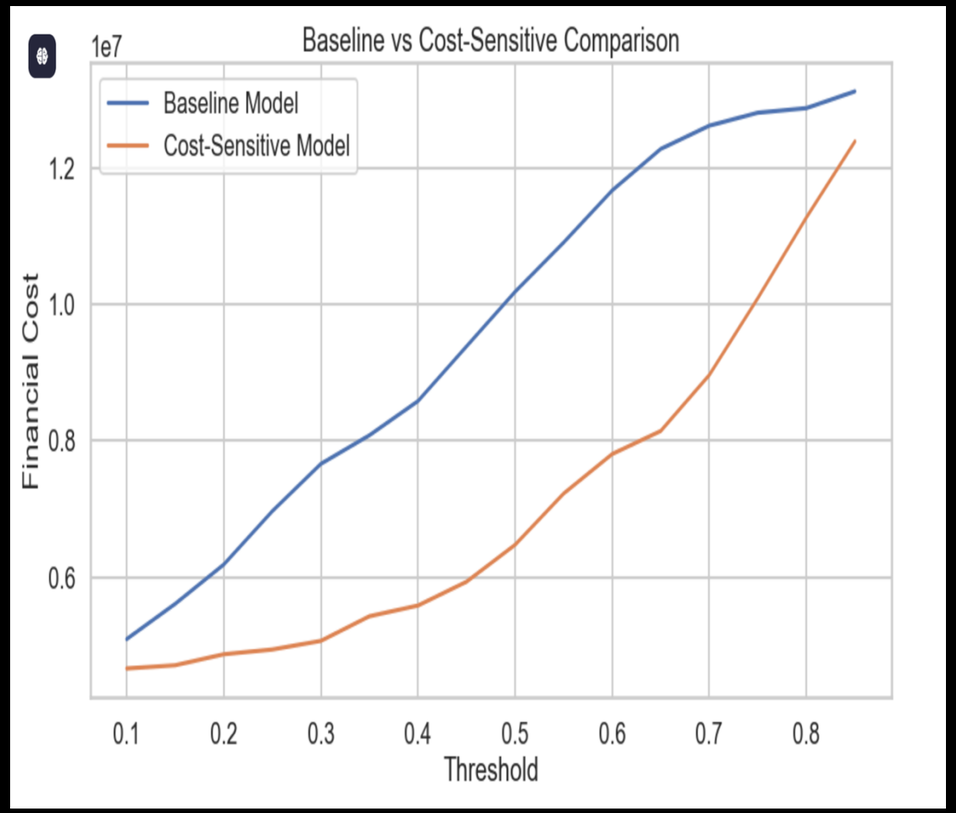
Increases recall and reduces false negatives by being more aggressive in flagging potential defaulters

Acceptable Trade-off

More false positives generated, but acceptable in banking context where missing a defaulter costs 10× more than a false alarm

Key Insight: In high-stakes financial applications, optimising for recall rather than accuracy delivers substantially better business outcomes, even with increased false positives.

Threshold Analysis: Baseline vs Cost-Sensitive Models



Financial cost was calculated across multiple decision thresholds ranging from 0.1 to 0.9 for both baseline and cost-sensitive Logistic Regression models.

Consistently Lower Costs

Cost-sensitive model produces lower financial cost at every threshold examined

Threshold Effect

As threshold increases, cost rises for both models but cost-sensitive remains superior

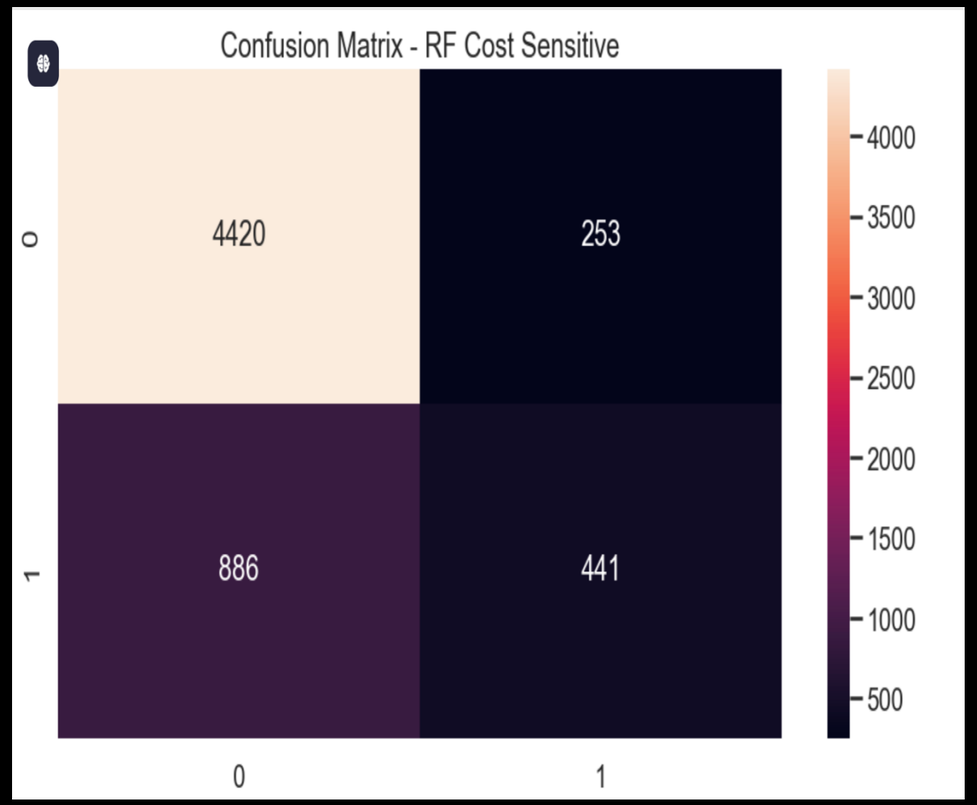
Stable Performance

Improved performance across all thresholds demonstrates alignment with financial objectives

Confusion Matrix: Error Breakdown Analysis

Key Findings

- True Negatives: 4,420 (correctly identified non-defaulters)
- False Positives: 253 (non-defaulters incorrectly flagged)
- False Negatives: 886 (defaulters missed by model)
- True Positives: 441 (correctly identified defaulters)



Critical Risk: 886 defaulters still missed represents ongoing financial risk requiring attention

Feature Importance: How It Was Derived

01

Model Selection

Random Forest model chosen for robust feature importance calculation

02

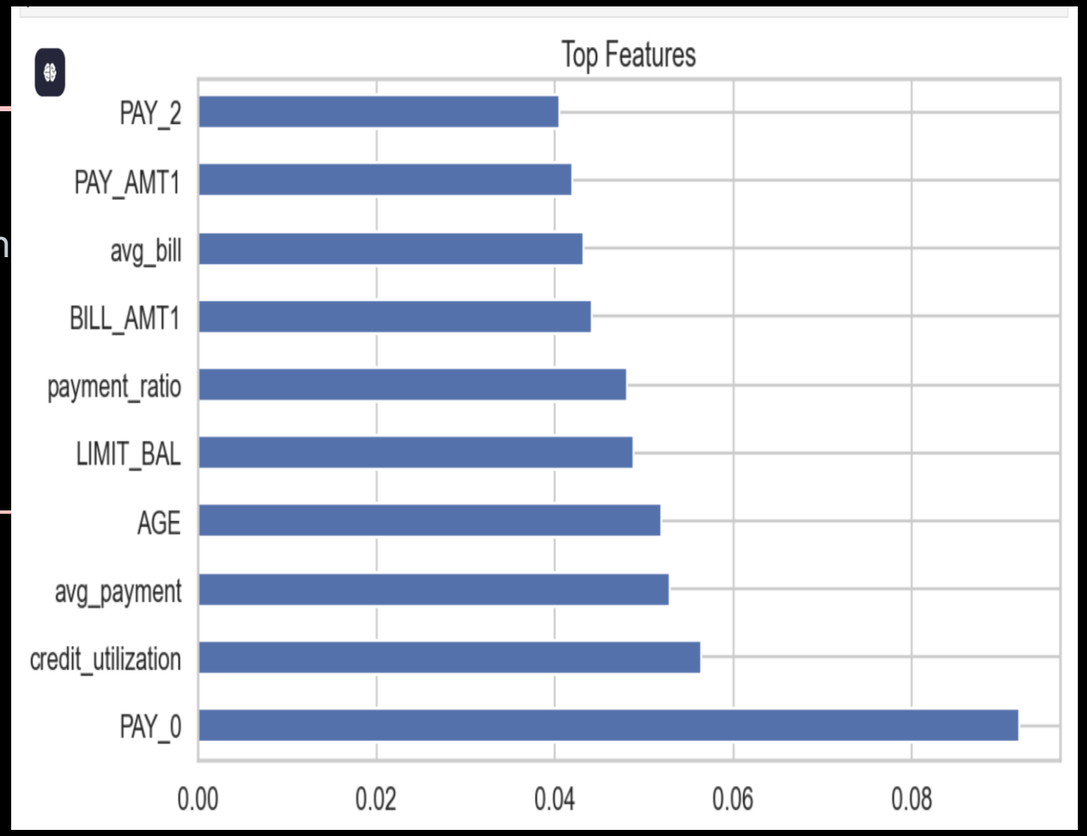
Split Contribution

Importance measured by contribution to decision splits across trees

03

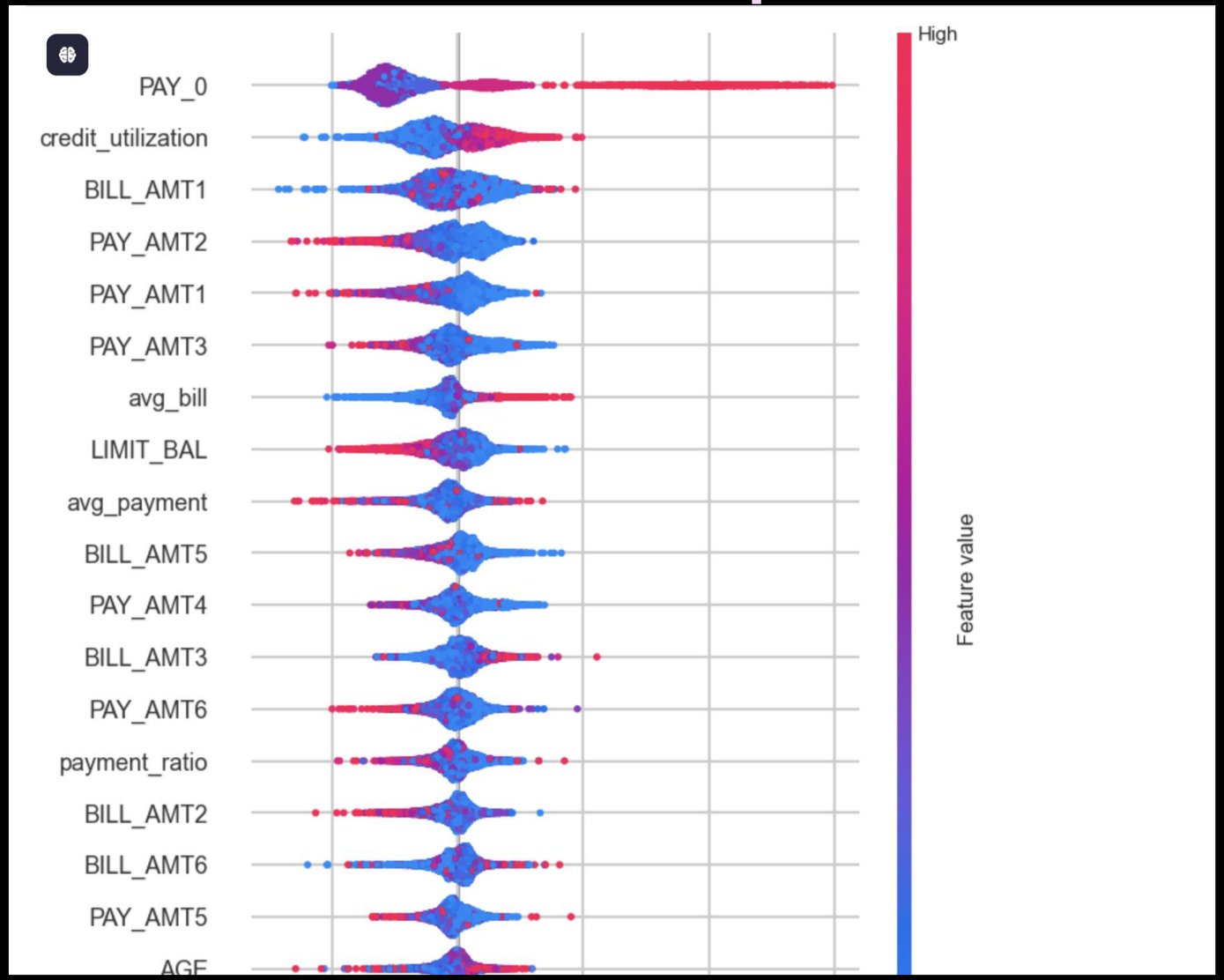
Ranking

Features ranked by total reduction in impurity across all splits



Key Result: PAY_0 emerged as most important feature. Recent repayment behaviour reflects current risk exposure more accurately than historical patterns, making it a powerful predictor of default probability.

SHAP Values: Interpretable Feature Impact



Methodology

SHAP (SHapley Additive exPlanations) values computed using XGBoost model to measure each feature's contribution to individual predictions.

Key Insights

- Confirms PAY_0 as most influential feature
- Shows direction of impact (positive/negative influence)
- Provides consistent interpretation across predictions

Key Insights from Analysis

Significant Recall Improvement

Cost-sensitive learning substantially improves recall, identifying more true defaulters whilst maintaining precision

Major Financial Loss Reduction

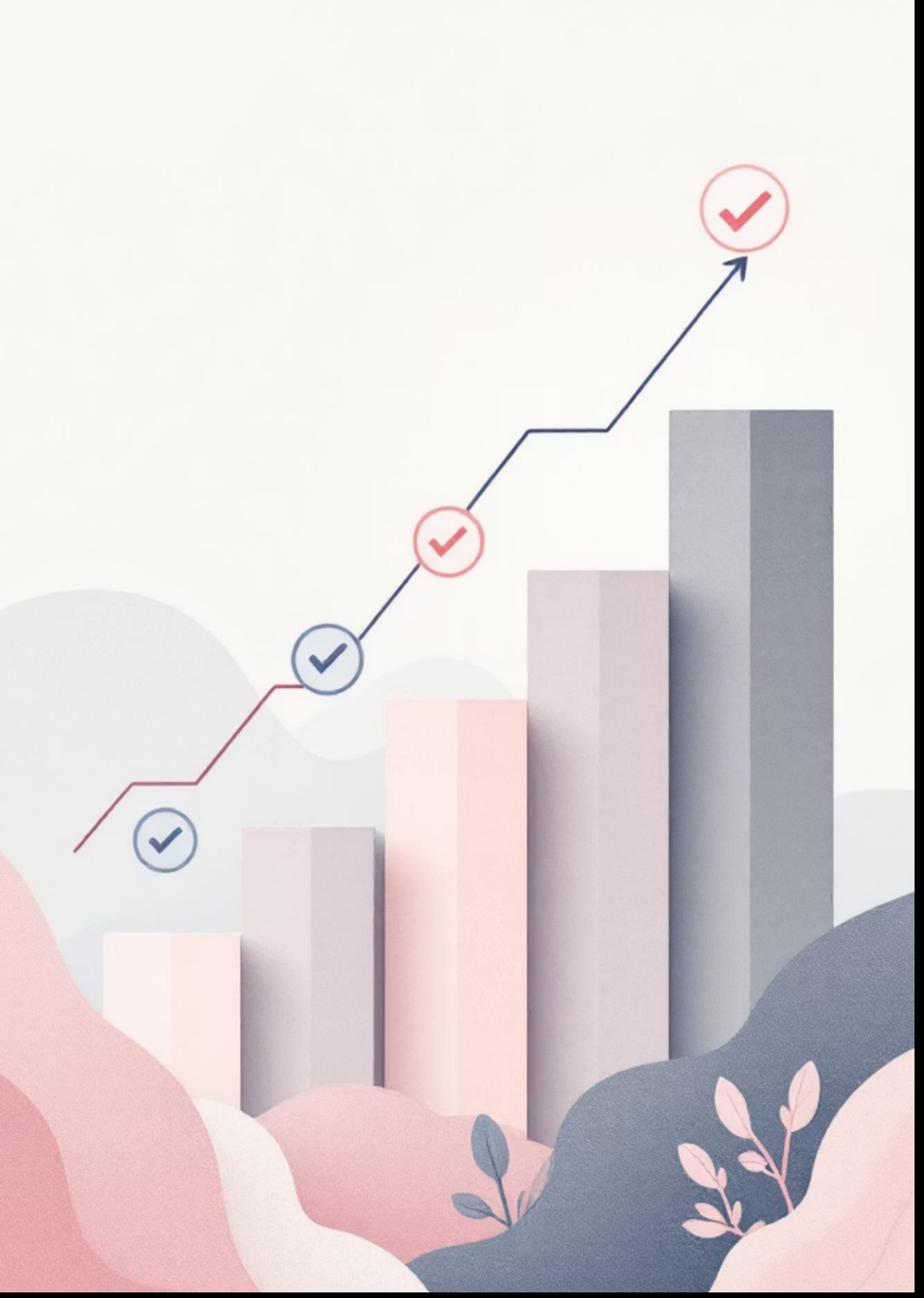
Financial loss decreased from ₹10.18 million to ₹4.66 million - a 54.2% reduction in exposure

Behaviour Features Dominate

Payment history and behavioural indicators drive predictions more than demographic or static features

Threshold Tuning Critical

Optimal threshold selection significantly impacts financial outcomes - requires careful calibration



Conclusion: Answering the Research Question

1

Cost-Sensitive Superiority

Cost-sensitive models demonstrably outperform standard models across all financial metrics

2

Risk Reduction

Significant reduction in financial risk through improved default prediction

3

Research Validation

Results clearly answer research question with statistical and financial evidence

Academic Sources

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